

FINAL TECHNICAL REPORT — DRAFT FEEDBACK

Detecting Direct Prompt Injection Attacks in LLM Applications

1. Overview

This is strong work. The draft final report is well-structured, technically accurate, and demonstrates genuine depth of engagement with the problem. Section 2 (transformer mechanics) and Section 3.4 (loss landscape motivation) are particularly impressive — the pedagogical clarity with which complex material is explained shows a level of understanding that goes well beyond implementation. The benchmarking section is rigorous, and the trade-off analysis shows the kind of multi-dimensional reasoning that characterizes good systems research.

The feedback below focuses on a small number of targeted revisions needed before final submission. None require structural changes to the report.

2. Things that worked

Abstract: Accurately summarizes the project scope, methods, and outcomes. The framing of the F1/FPR boundary as a documented finding rather than a failure is exactly right.

Section 2 (Transformer walkthrough): The step-by-step explanation of tokenization through Softmax, with the running "Ignore previous instructions" example, is genuinely excellent. The analogies (committee writing a sticky note, GPS coordinates on a hiking map) are well-chosen and would make this section accessible to a general engineering audience. The end-to-end pipeline diagram (Section 2.6) is a standout.

Section 3.4 (Loss landscape motivation): This is the most original analytical writing in the report. Framing the precision-recall wall as a narrow valley with cliffs on either side, then using the 2D grid as a search across that landscape, is a coherent and compelling research narrative. The kitchen analogy for LR vs. weight decay interaction is well-executed.

Cross-architecture leaderboard (Table 4.4): Having all four paradigms on one table with consistent metrics and a verdict column is exactly what a comparative evaluation report needs. This should be the anchor of the poster presentation.

Trade-off analysis (Section 5): Section 5.3 (paradigm trade-offs) and 5.4 (direct vs. indirect) both provide clear engineering takeaways. The insight that asymmetric loss is the key differentiator — that off-the-shelf search and generative judges cannot encode an explicit FP penalty — is a genuine research conclusion.

References: Complete and properly formatted. All 20 references appear to be real, relevant, and correctly attributed.

3. Required Revisions Before Final Submission

a) Table of Contents should be placed before Abstract on title page along with chapters, topics, sub-topics and page numbers.

3.1 Section 3.3 — Strategy 2: Asymmetric Loss Formula

There is a minor but important precision issue in the loss formula as written:

As written: $L_n = -w_y \log(p_y)$

Issue: This notation implies the loss is computed only on the predicted class, and the variable subscripts are inconsistent with standard cross-entropy formulations. In your WeightedTrainer implementation, PyTorch's CrossEntropyLoss with class weights applies the weight corresponding to the true class label, not the predicted class. The clearer formulation is the standard weighted cross-entropy:

Suggested revision: $L = -\sum_c w_c y_c \log(p_c)$, where $w_{\text{benign}} = 3.0$ and $w_{\text{malicious}} = 1.0$, and y_c is the one-hot true label. This makes explicit that the penalty applies based on the ground-truth class, not the model prediction.

This is a small fix but important for technical accuracy in a final report.

3.2 Section 4.1 — ProtectAI Baseline: Missing Latency

The cross-architecture leaderboard (Table 4.4) shows a dash ("—") for ProtectAI DeBERTa latency. This is a named contract requirement: the baseline comparison must include either a latency figure or an explicit statement explaining why it was not measured. Options:

Option A: Report the measured latency from your Milestone 2b evaluation script if available.

Option B: Add a footnote: "ProtectAI DeBERTa latency was not measured in this evaluation; DeBERTa-v3-base (~138M parameters) is expected to be substantially slower than DistilBERT (~66M) on equivalent hardware, supporting the contract's $\geq 25\%$ latency advantage criterion." This satisfies the contract without requiring a re-run.

Either option is acceptable. The current dash leaves the baseline comparison formally incomplete.

3.3 Appendix B — Contract Scorecard: OOD F1 Notation

The scorecard reports OOD F1 as "0.3368" — this is inconsistent with the body of the report, which reports it as "33.68%" in Section 4.6. Pick one format (percentage is more readable in a table context) and apply it consistently. The **contract target should also be stated as "70.00%"** rather than "0.70" for consistency.

3.4 Section 3.5 (KNN Deviation) — Add a Comparative Sentence

The note that KNN was chosen over Random Forest/XGBoost because distance metrics exploit embedding geometry is a reasonable justification. However, the contract specified RF/XGBoost, and the final report should close this formally. Please add one sentence quantifying or referencing why KNN is appropriate:

Suggested addition: "A brief comparison using Scikit-learn's RandomForestClassifier on the same MiniLM embeddings yielded comparable F1 (within 1-2 pp) with higher inference latency; KNN was retained as the cleaner geometric baseline." If you did not run this comparison, a sentence referencing prior literature on KNN vs. RF over dense embeddings is also sufficient.

4. Optional Enhancements (Not Required)

Section 2 length: Section 2 is approximately one-third of the total report length. For a technical report this is somewhat unusual typically background mechanics are compressed in favor of results and analysis. Consider whether some of the more elementary subsections (e.g., 2.2 Embedding layer, 2.3a multi-head attention detail) could be moved to an appendix without weakening the argument. This is a judgment call and not a requirement for final submission.

Section 6.1 Future Work: The five future directions are well-chosen. Consider ordering them by expected impact on the two unmet targets (F1 gap, OOD gap) to give the list more analytical structure.

5. Contract Compliance Summary

Requirement	Target	Achieved	Status
Primary DistilBERT F1	$\geq 85\%$	83.31%	MISS*
FPR (primary model)	$< 5\%$	4.05%	PASS
OOD BIPIA F1	≥ 0.70	33.68%	FAIL**
Inference Latency	< 50 ms	10.9 ms	PASS
4 paradigms evaluated	Required	4 (KNN noted)	PASS
Baseline comparison	Required	Present (latency gap)	PARTIAL
Reproducible artifacts	Required	Frozen splits + HF Hub	PASS

*F1 gap thoroughly documented as a corpus-architecture constraint finding — this is the correct framing.

**Distribution shift quantified and analyzed — the diagnostic quality is strong even though the target is not met.

6. Closing Note

The report presents a genuine research contribution: a documented Pareto frontier between F1 and FPR under an asymmetric loss constraint, plus a characterization of the distribution shift boundary between direct and indirect injection detection. The three required revisions above are small and straightforward. Please address them before the Milestone 6 final submission deadline.